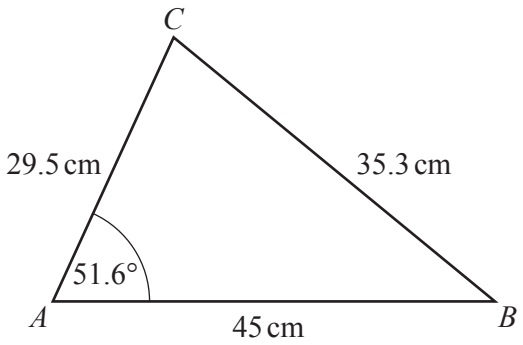
0580_m22_qp_42_Q4

4 **(a)**



NOT TO SCALE

In triangle ABC, $AB = 45$ cm, $AC = 29.5$ cm, $BC = 35.3$ cm and angle $CAB = 51.6°$.

(i) Calculate angle ABC.

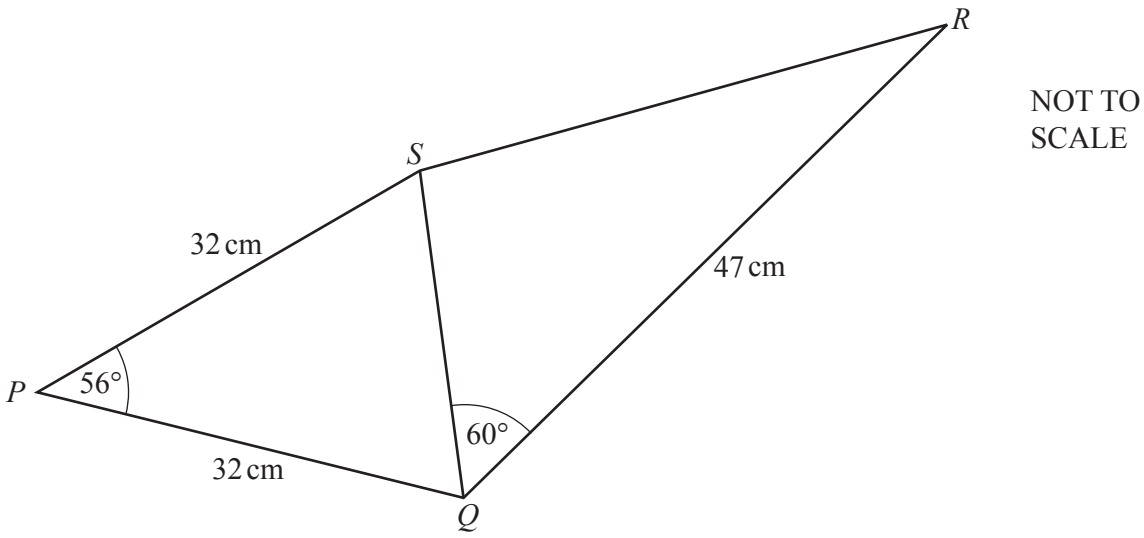
Angle ABC = ... [3]

(ii) Calculate the area of triangle ABC.

..... cm^2 [2]

0580_m22_qp_42_Q4

(b)



The diagram shows a quadrilateral *PQRS* formed from two triangles, *PQS* and *QRS*.
Triangle *PQS* is isosceles, with $PQ = PS = 32$ cm and angle $SPQ = 56°$.
$QR = 47$ cm and angle $SQR = 60°$.

(i) Calculate *SR*.

SR = ... cm [4]

(ii) Calculate the shortest distance from *P* to *SQ*.

..... cm [3]